

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 - Scenario-Based Questions

Course Code: DSA0405 Course Title: FUNDAMENTALS OF DATA SCIENCE

CO Assessed: CO4 – Scenario-Based Questions Assessment: Assessment Tool 1– Scenario Based Questions

Weightage: 40% of CIA Total Marks: 100

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Section / Batch:	AIDS	Date:	20-08-2026

Code of Conduct

I M.Mohamed Bazith (192472052) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Bazith

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

QUESTIONS :

6. ONLINE LEARNING PLATFORM

Topic: Confidence Interval

An online learning platform measures the completion time of a course for 100 students. The mean completion time is 14 days and the standard deviation is 3 days. Construct a 95% confidence interval for the true mean completion time.

7. HOSPITAL PATIENT SATISFACTION

Topic: Confidence Interval

A hospital surveys 400 randomly selected patients, and 340 report that they are satisfied with the service. Construct a 95% confidence interval for the proportion of satisfied patients.

8. TELECOM DATA USAGE

Topic: Confidence Interval

A telecom company wants to estimate the average monthly data usage of its customers. A sample of 225 customers has a mean usage of 16 GB and a standard deviation of 5 GB. Construct a 99% confidence interval and interpret the result.

9. CUSTOMER SUPPORT RESOLUTION TIME

Topic: Hypothesis Testing Using CI

A company claims that the average time taken by its customer-support team to resolve a complaint is 24 hours. A sample produces a 95% confidence interval of (25.2, 27.8) hours. Based only on this confidence interval, would you reject the company's claim? Explain.

10. MANUFACTURING – PACKAGED PRODUCT WEIGHT

Topic: Hypothesis Testing Using CI

A manufacturer claims that the average weight of a packaged product is 500 grams. A 95% confidence interval based on a sample is (496, 502) grams. Is the claimed value of 500 grams consistent with the sample evidence? Explain your answer.

ANSWERS:

6. ONLINE LEARNING PLATFORM – CONFIDENCE INTERVAL

Given Information

Sample size, $n = 100$

Sample mean, $\bar{x} = 14$ days

Standard deviation, $\sigma = 3$ days

Confidence level = 95%

1. Compute the Standard Error (SE)

$$SE = \sigma / \sqrt{n} = 3 / \sqrt{100} = 3 / 10 = 0.3$$

2. Apply the Confidence Interval Formula

$$CI = \bar{x} \pm (z \times SE), \text{ where } z = 1.96 \text{ for } 95\% \text{ confidence}$$

$$\text{Margin of Error} = 1.96 \times 0.3 = 0.588$$

$$CI = 14 \pm 0.588 = (13.412, 14.588) \text{ days}$$

3. Conclusion

We are 95% confident that the true mean completion time of the course lies between 13.41 days and 14.59 days.

Industry-Oriented Conclusion

The online learning platform can be confident that most students complete the course in approximately 13 to 15 days on average. This information can help the platform plan course schedules, set realistic completion deadlines, and design timely reminders or support interventions for students who are falling behind.

7. HOSPITAL PATIENT SATISFACTION – CONFIDENCE INTERVAL

Given Information

Sample size, $n = 400$

Number satisfied, $x = 340$

Sample proportion, $\hat{p} = 340 / 400 = 0.85$

Confidence level = 95%

1. Compute the Standard Error (SE)

$$SE = \sqrt{[\hat{p}(1 - \hat{p}) / n]} = \sqrt{[(0.85 \times 0.15) / 400]} = \sqrt{0.00031875} \approx 0.01785$$

2. Apply the Confidence Interval Formula

CI = $\hat{p} \pm (z \times SE)$, where $z = 1.96$ for 95% confidence

Margin of Error = $1.96 \times 0.01785 \approx 0.035$

CI = $0.85 \pm 0.035 = (0.815, 0.885)$, i.e., (81.5%, 88.5%)

3. Conclusion

We are 95% confident that the true proportion of satisfied patients lies between 81.5% and 88.5%.

Industry-Oriented Conclusion

The hospital can conclude that a large majority of its patients (roughly 82% to 89%) are satisfied with the service. This is a positive indicator of service quality, though the hospital may continue monitoring feedback to identify and address the concerns of the remaining unsatisfied patients.

8. TELECOM DATA USAGE – CONFIDENCE INTERVAL

Given Information

Sample size, $n = 225$

Sample mean, $\bar{x} = 16$ GB

Standard deviation, $\sigma = 5$ GB

Confidence level = 99%

1. Compute the Standard Error (SE)

$SE = \sigma / \sqrt{n} = 5 / \sqrt{225} = 5 / 15 \approx 0.333$

2. Apply the Confidence Interval Formula

CI = $\bar{x} \pm (z \times SE)$, where $z = 2.576$ for 99% confidence

Margin of Error = $2.576 \times 0.333 \approx 0.859$

CI = $16 \pm 0.859 = (15.14, 16.86)$ GB

3. Interpretation

We are 99% confident that the true average monthly data usage of all customers lies between 15.14 GB and 16.86 GB.

Industry-Oriented Conclusion

The telecom company can use this range to plan network capacity and design data plans that suit typical customer usage (around 15–17 GB per month). This helps in forecasting infrastructure needs and creating appropriately priced data packages.

9. CUSTOMER SUPPORT RESOLUTION TIME – HYPOTHESIS TESTING USING CI

Given Information

Company's claim: Average resolution time = 24 hours

95% Confidence Interval = (25.2, 27.8) hours

1. State the Hypotheses

$H_0: \mu = 24$ hours (the claim is true)

$H_1: \mu \neq 24$ hours (the claim is false)

2. Compare the Claim with the Confidence Interval

The claimed value of 24 hours does not lie within the interval (25.2, 27.8).

3. Decision

Since the claimed value falls outside the 95% confidence interval, we reject the company's claim at the 5% significance level.

4. Conclusion

There is sufficient statistical evidence that the true average resolution time is greater than 24 hours, since the entire interval (25.2 to 27.8 hours) lies above the claimed value.

Industry-Oriented Conclusion

The company's claim of a 24-hour resolution time is not supported by the data. Actual resolution times appear to be higher (roughly 25 to 28 hours), so the company should investigate its support process and take corrective measures such as additional staffing, training, or process automation to reduce resolution time.

10. MANUFACTURING – PACKAGED PRODUCT WEIGHT

Given Information

Manufacturer's claim: Average weight = 500 grams

95% Confidence Interval = (496, 502) grams

1. State the Hypotheses

$H_0: \mu = 500$ grams (the claim is true)

$H_1: \mu \neq 500$ grams (the claim is false)

2. Compare the Claim with the Confidence Interval

The claimed value of 500 grams lies within the interval (496, 502).

3. Decision

Since the claimed value falls inside the 95% confidence interval, we fail to reject the null hypothesis at the 5% significance level.

4. Conclusion

The claimed average weight of 500 grams is consistent with the sample evidence. There is not enough statistical evidence to say that the true average weight differs from 500 grams.

Industry-Oriented Conclusion

The manufacturer's claim about the packaged product weight appears accurate and reliable based on the sample data. The company can continue its current production process, while still performing routine quality checks to maintain consistency.